

[Quiz—4] Spring 2024: CSE 309: Theory of Automata

Tuesday, May 7, 2024. Total Marks: 10, Duration: 15 minutes

— SOLUTION —

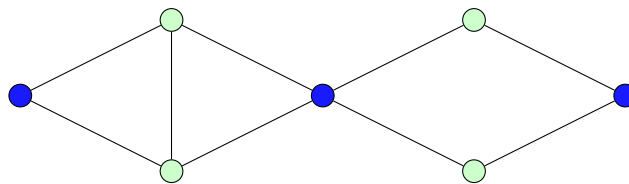
Question 1

[10 points]

Show that the problem IS (independent set problem) is in the class **NP**. We define the problem as follows: Given a simple graph $G = (V, E)$ and an integer k , does there exist a subset $S \subseteq V$ such that $|S| \geq k$ and no two vertices in S are adjacent in G ?

$$\text{IS} = \{ \langle G, k \rangle \mid G \text{ is a graph and there exists an independent set } S \text{ of size } k \text{ in } G \}.$$

For example, in the following graph blue vertices form an independent set of size 3.



Solution: To show that IS is in **NP**, we need to show that given a certificate S (a subset of vertices of size k), we can verify in polynomial time that S is an independent set in G .

To verify that S is an independent set, we need to check that no two vertices in S are adjacent in G . This can be done by checking all pairs of vertices in S and verifying that there is no edge between them. This verification can be done in $O(k^2)$ time, which is polynomial in the size of the input.

Therefore, IS is in **NP**.